

Roll no: 26k-0006.

Ques no 6 (Smart EV Charging and Parking Management System)

Algorithm:

- 1) Start
- 2) Read Vehicle-Type (E=Electric, H=Hybrid), Current SOC (Battery percentage), Required SOC (Target battery percentage), Duration (Expected stay duration in hours), Current-Time (24-hour format), Club-Member (Y/N), Disabled-Priority (Y/N) and Station-Available (Y/N)
- 3) Check Availability & Eligibility Rules:
 - If Station-Available == 'N'
 - If Vehicle-Type == 'H' Display "Charging unavailable - Parking only."
 - Else Display "No charging slot available." print "Vehicle does not qualify for EV charging."
- 4) Calculate Charging Needs:
 - Required-Charging = Required-SOC - Current-SOC
 - If Required-Charging ≤ 0
Display "No charging required." set Charging-Cost and move to step 7.
- 5) Assign Priority Categories.
 - Priority 1 (Emergency Charging): IF Current-SOC ≤ 15 AND Required-Charging ≥ 80 . Display "Priority 1 Emergency Charging"

Roll no: 26k-0006

- Priority 2 (Priority Customer): If Disabled-Priority == 'Y' OR Club-Member == 'Y', AND Current-SOC < 30. Display "Priority 2: Priority Customer".

- Priority 3 (Normal Charging) For all other cases requiring juice. Display "Priority 3: Normal Charging".

6) Calculate Base Charging Cost.

- Based on Current-Time

- Peak Hours (17 to 22 / 5 PM to 10 PM): Base-Rate = 50 per unit.

- Off Peak Hours (Before 17 or after 22): Base-Rate = 35 per unit.

- Calculate Charging-Cost = Required Charging * Base-Rate
- If Off-Peak hours match AND Club-Member == 'Y', apply a 20% discount on the charging amount.

7) Calculate Parking Fee Rules:

- If Disabled-Priority == 'Y', Parking-Fee = 0.

- Else

- If Duration <= 2: Parking Fee = 200

- If Duration > 2: Parking Fee = 400

- If Duration > 5: Parking Fee = 700

Roll no: 26k-0006

- 8) Apply Special Customer Discount
 - If Disabled Priority == 'Y' OR Club-Member == 'Y':
Apply an additional discount 20% onto the calculated Parking-Fee.
- 9) Verify Stay Duration warnings
 - If Duration > 8: Set Warning-Message = "Long-stay warning: Please relocate your vehicle after charging."
 - Else: Set Warning-Message = "Standard parking duration."
- 10) Final payable = Charging-cost + Parking-Fee
- 11) Display System Output Profile Summary Panel.
- 12) End the program.

PAC:

Given Data	Processing Required	Output Requirements
• Vehicle-Type(E/H)	• Calculate charging deficit.	• Alert (No charging slot available)
• Current-SOC/ Required-SOC	• Classify 3 priority levels.	• Tier classification
• Duration	• Branch pricing parameters using 24-hr time filters (Peak vs Off Peak)	• output tracking category metrics.
• Current-time	• Layer cumulative	• Receipt grid
• Flags: Club-Member, Disabled Priority, Station Available.	• conditions for custom group reductions.	• printout detailing costs, discounts, and final-balance due.
	• Track duration.	

Roll no: 26k-0006

IPO chart:

Input	Processing	Output
• Vehicle type • Charging level • Operational system variable flags.	• Filter out non-eligible vehicle. • Calculate target power requirement amounts. • Determine unit base rate. • Compute parking structures.	• Priority categorization classification. • Itemized parking alerts, discounts, and total currency balance requirements.

Pseudocode:

BEGIN

DECLARE Vehicle-Type, Club, Disabled, Available
AS CHARACTER

DECLARE Current-SOC, Required-SOC, Duration,
Current-Time AS INTEGER

DECLARE Req-Charging, Base-Rate, Charging-
Cost, Parking-Fee, Final-Payable AS REAL.

DECLARE Priority-Msg, Warning-Msg AS STRING

INPUT Vehicle-Type, Current-SOC, Required-SOC,
Duration, Current-Time, Club, Disabled, Available.

IF Available == 'N' THEN

Roll no: 26k-0006

```
IF Vehicle-Type == 'H' THEN  
    PRINT "Charging unavailable- Parking only".  
    Charging-cost = 0  
    GOTO Parking-SECTION
```

```
ELSE  
    PRINT "No charging slot available"  
    EXIT
```

```
ENDIF
```

```
ENDIF
```

```
Req-charging = Req-SOC - Current-SOC
```

```
IF Req-charging <= 0 THEN
```

```
    Charging-cost = 0
```

```
    Priority-Msg = "No charging Required".
```

```
ELSE
```

```
    IF (Current-SOC <= 15 AND Req-charging >= 80) THEN
```

```
        Priority-Msg = "Priority 1- Emergency Charging"
```

```
        IF (Current-Time >= 17 AND Current-Time <= 22)
```

```
            THEN Base-Rate = 50 ELSE Base-Rate = 35
```

```
        ENDIF
```

```
        Charging-Cost = Req-charging * Base-Rate
```

```
    ELSE
```

```
        IF (Disabled == 'Y') OR (Club == 'Y') AND (Current-  
        SOC < 30) THEN Priority-Msg = "Priority 2 -  
        Priority Customer"
```

```
    ELSE
```


Roll no: 26k-0006

Priority-Msg = "Priority 3 - Normal Charging"
ENDIF

IF (Current-Time $\geq$ 17) AND (Current-Time $\leq$ 22) THEN

Base-Rate = 50

Charging-Cost = Req-Charging * Base-Rate
ELSE

Base-Rate = 35

Charging-Cost = Req-Charging * Base-Rate

IF Club == 'Y' THEN Charging-Cost =

Charging-Cost * 0.80 ENDIF

ENDIF

ENDIF

ENDIF

IF Disabled == 'Y' THEN Parking-Fee = 0

ELSE IF Duration $\leq$ 2 THEN Parking-Fee = 200

ELSE IF Duration $\leq$ 5 THEN Parking-Fee = 400

ELSE Parking-Fee = 700.

ENDIF

IF (Club == 'Y') OR (Disabled == 'Y') THEN

Parking-Fee = Parking-Fee * 0.80.

ENDIF

ENDIF

IF Duration $>$ 8 THEN Warning-Msg = "Long-stay
warning"

Roll no: 26k-0006

ELSE

Warning-Msg = "Standard Parking Duration"

ENDIF

Final-Payable = charging-Cost + Parking-Fee

PRINT "Priority Group Status", Priority-Msg

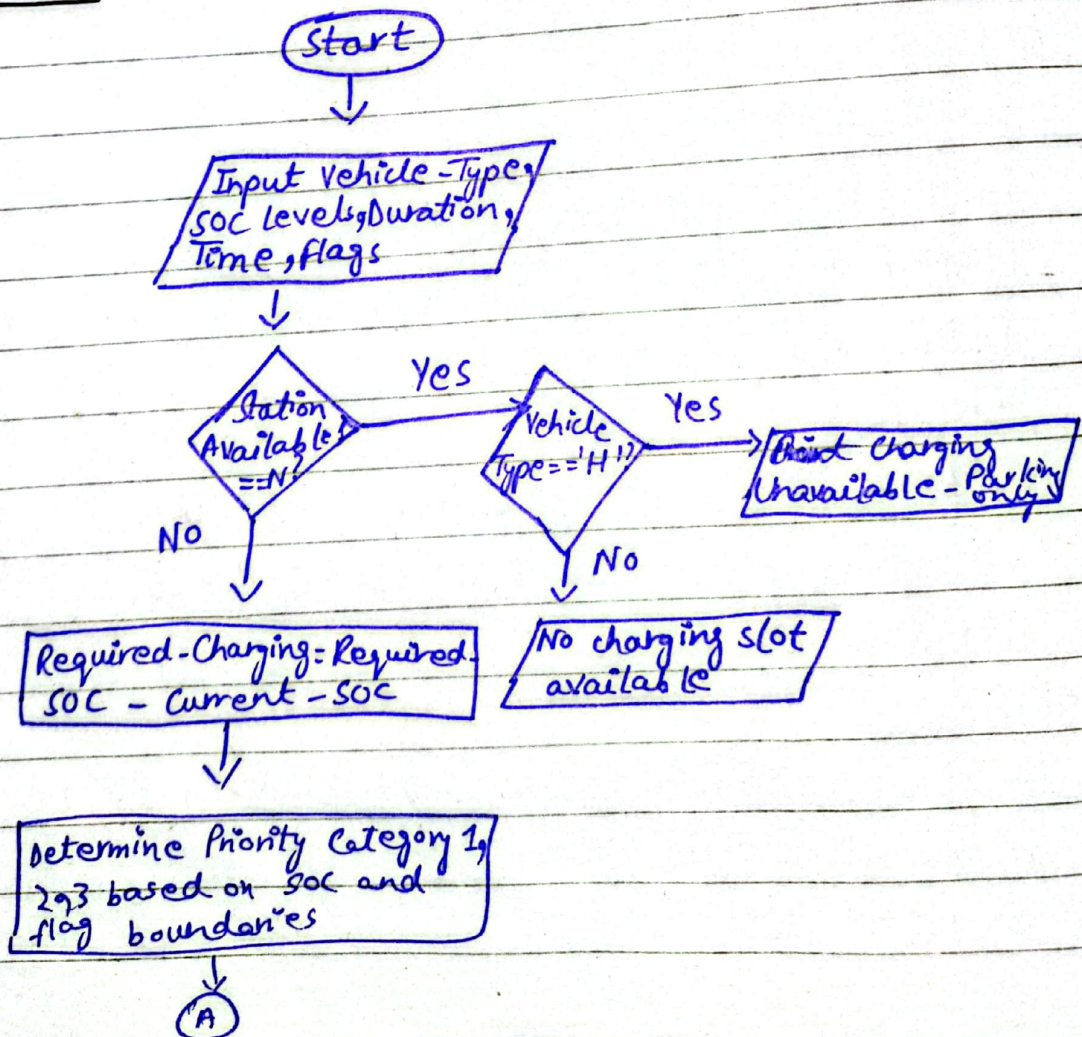
PRINT "Total Cumulative Invoice Summary Bill, Rs",

Final-payable.

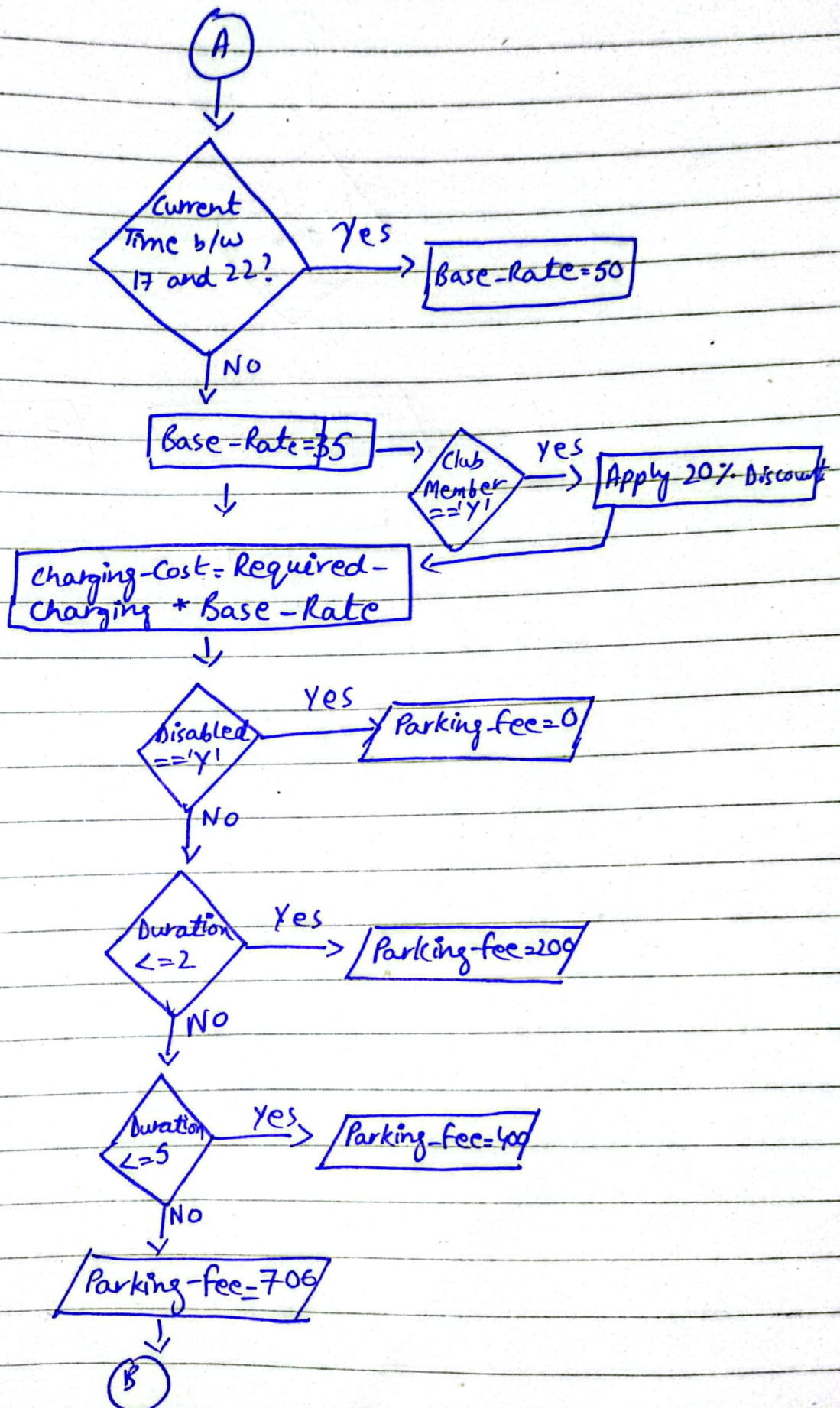
PRINT "Active Directives: ", Warning-Msg

END

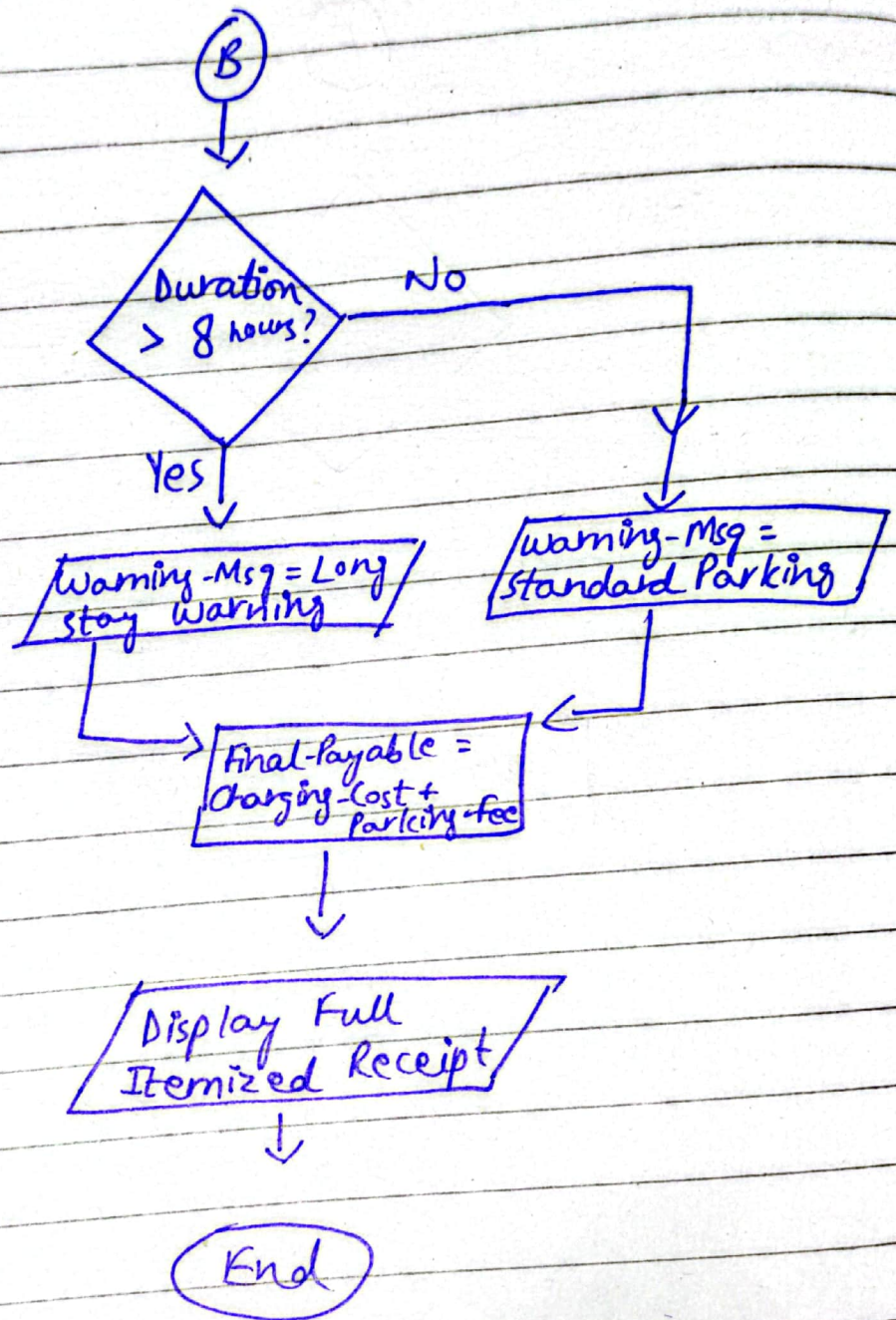
Flowchart:



Roll no: 26k-0006



Roll-no: 26k-0006



— X —